

Finding Picking Point Using Straight Line Detection of Label Attachment Machine

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Abstract— We developed a new technique for detecting the position of labels using a SCARA Robot with the end-effector vacuum pad. The technique first found the image edge with a Canny detector, then selected straight lines and their intersections. It assumed that the label was basically rectangular in shape and roughly aligned so that the key edges were close to horizontal and vertical directions. The slope of the horizontal edge defined the rotational of the label. Otherwise, reducing the image tilting and the distortion by using a perspective transform matrix and thereafter converting the image coordinate into the SCARA robot coordinate. Through the factory environment, the experiment is conducted using various captured image conditions under a lighting source. The performance of the system provides the acceptable accuracy all of the picking coordinates with positioning errors, which x-coordinate are 0.01 – 0.25 mm, y-coordinate are 0.01 – 0.32 mm and rotation angle are 0.05° – 3.94°. In trials, the SCARA robot was able to grip the label precisely in each test.

Keywords— *SCARA robot, Robot positioning, Straight line detection, Perspective Transform.*

I. INTRODUCTION

In the last decade, the number of competitions for robotic arm design in factory automation has increased extensively. Performance has also increased significantly. However, many types of robots that have different functions. For example, Selective Compliance Assembly Robot Arm (SCARA) robots are lightweight and have small footprints, making them ideal for applications in crowded spaces. They are capable of very fast cycle times and are also suitable for pick and place tasks [1]. Machine vision is the substitution of human visual sense and judgment capabilities with a video camera and computer to perform an inspection task [2]. It is the automatic acquisition and analysis of images to obtain desired data for controlling or evaluating a specific part or activity [3]. Robotic arms using machine vision techniques have been used wisely in industry for the location and orientation of the object, then commanding the robot device to the expected coordinates to grip the workpiece.

Many studies has discussed robotic pick and place systems. Nerakaea, Uangpairoj, and Chamniprasart designed a prototype of a flexible automatic assembly system combined with a machine vision system. Using LabVIEW NI vision software and NI vision builder, the captured image was used to measure the exact target position for placing the assembly part. They calibrated their system by using image, which has a size standard of circle Forty-eight circles (6x8 rows) with 10 mm diameter in its application [4]. Huang, Lin, and Chen used a scheme to measure the relative distance between the object

and robot arm using edge detection in captured images [5]. Its process involves the environment indoor and hall, and the color differentiating of input visualization from the camera to locate the candidate block of fitting color requirements and thereafter define the sector. Kim et al [6] described vision based bin picking system using multiple local features for an industrial robot. The setting of the working environment for bin-picking, the detection of object location, pose estimation, distance measurement and surface orientation angle detection steps were included in their system. Pyo et al [7] described a vision-based angle estimation algorithm for bin picking. They described a rotation angle estimation method, which was not influenced by the pose of the object and used an affine transformation to preprocess the image for invariance to object posture. Hu et al. presented a vision-based SCARA system for spanner fetching and placing, which can be applied in traditional factories [8]. They used a SCARA robot with a 500 mm working range, and four degrees of freedom. It was designed to be low-cost. Farag, Abd Ghafar and Alsibai [9] aimed to increase the impact of computer vision on robotic localization and grasping applications. They implemented comparison of edge detect methods, object detection and localization to enable robotic grasping and positioning using SCARA. However, the entire working environment in the factory has different lighting conditions to achieve better precision of vision processing.

II. OUR METHOD

A. Overview

There are two steps in Label attachment, the label barcode must be transferred the data to the server and the label attached to a printed circuit board (PCB), which has a minimum label size of 6×6 mm. There are some errors in manual processes, - missing scanned barcode data to be sent to the server and wrong alignment of label when attaching it, because of low visibility of the labeling box as shown in figure (3e). Moreover, the production investment cost in the manpower is raised for the long-term future. Robotic systems can automate processes to reduce human lack of attention during work, reduce cycle and lead time, and expenses. Automated machines can achieve better performance and high precision in production lines, which led to a desire to develop the label attachment machine. Here, we focused on finding the positions of labels, using simple fast image processing, to design software to integrated with a SCARA robot.

The basic hardware for label localization [10] contained four components, - a robot, industrial camera, lighting source, and industrial personal computer. The robot arm was a Delta

SCARA robotics model of DRS60L6SS1BN002: the single camera was Basler model 14GM-2500AC, with 50 mm lens, resolution size of 5M pixels, and 14 fps, - seen in the acrylic box above and at the right-hand of the robot. The SCARA robot performs the picking and attaching object operation. An added light source maintained stable lighting for a captured images and adapting to the factory environment. The height from camera to the target object was ~750 mm, allowing space for the robot to pick the object comfortably. A 5.5 mm vacuum pad was attached to the end effector of the robot. Images were captured and processed on the industrial computer, Nuvo-6032 Window 7 Intel Skylake, with 6th Gen core. The system is shown in Figure 1.

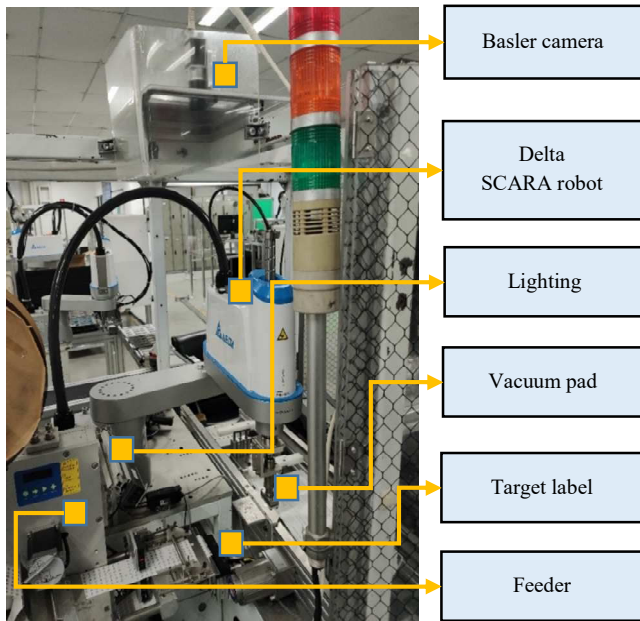


Fig. 1. Hardware device systems

B. Algorithm for finding label position

The positioning algorithm was programmed by EMGUCV [11], a cross-platform .Net wrapper to the OpenCV [12], an image processing library, which captured the image from the Programmable Logic Controller (PLC) via TCP/IP. The PLC was an intermediary for communicating and controlling output devices that was programmed by a ladder diagram, and stored various data, including labels, status values, etc. The label coordinate is defined through the data flow as shown in figure 2.

An image was captured, transformed the image to the grayscale and conduct the edge detection [13] image. Then the seven labels in each image was detected. If they were not detected, a status message was sent to the PLC “no label found”. Then we search for the next label. The label detection algorithm, finding the left edge as a vertical line and the bottom edge as a horizontal line using Hough transform. Intersecting both vertical and horizontal lines make a “gripping point” to transform the image coordinates to robot coordinates. The label angle was determined from the horizontal line. Then the label position was sent to the robot, and responding to the PLC with a “label found” message.

III. EXPERIMENTS

In this section, the method used is explained in detail about detecting the labels, localization and orientation of picking

coordinates, image and robot coordinate transformation and experiment results.

A. Detecting the Labels

The captured 2592×1944 pixel image was converted to grayscale -see figure (3a), where the label was on the bar of the feeder. Our approach used an OpenCV GaussianBlur function with kernel size of 5×5, a Canny [14] edge detector and detected intersections between the outer edges of the label to find the picking point.

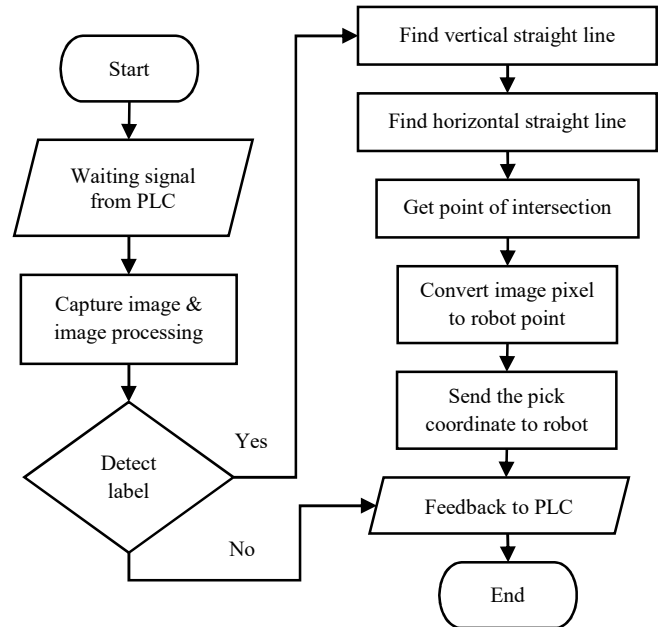


Fig. 2. Algorithm for finding the label position

There are three steps to find picking points in the grayscale image. First, a canny edge detector with 5×5 kernel was applied to find out approximate shapes of objects. Then an ROI for the first label was set as input to eliminate some extraneous noise and speed up computation - see the red rectangle in figure (3b). Known differences to the subsequent labels were used to set ROIs for them. Inside these ROIs, the OpenCV FindContour function was used to find potential contour: the moments of these contours identified a rectangle of suitable size and its center point was set the reference to point of the labels. Noise inside the label was filled up as black to remove some undesired scattered white. Figure (3c) shows the image after removing noise. Finally, straight lines were detected by a Hough transform [15] - finding vertical and horizontal lines from the edges of images. The intersection points between vertical and horizontal lines provided the picking point, while the angle obtained from the horizontal line, see figure (3d).

B. Localization and orientation of picking coordinates

Both labels and the vacuum suction attachment are rectangular in shape, where the picking point is at the left bottom corner of labels.

Given two lines, L1 and L2, in two-dimensional space [16], where line L1 is defined by points (x_1, y_1) and (x_2, y_2) , and line L2 is defined by (x_3, y_3) and (x_4, y_4) , the intersection (x, y) is given by equation (1). The orientation of the label, given by the horizontal line L2, is computed by equation (2).

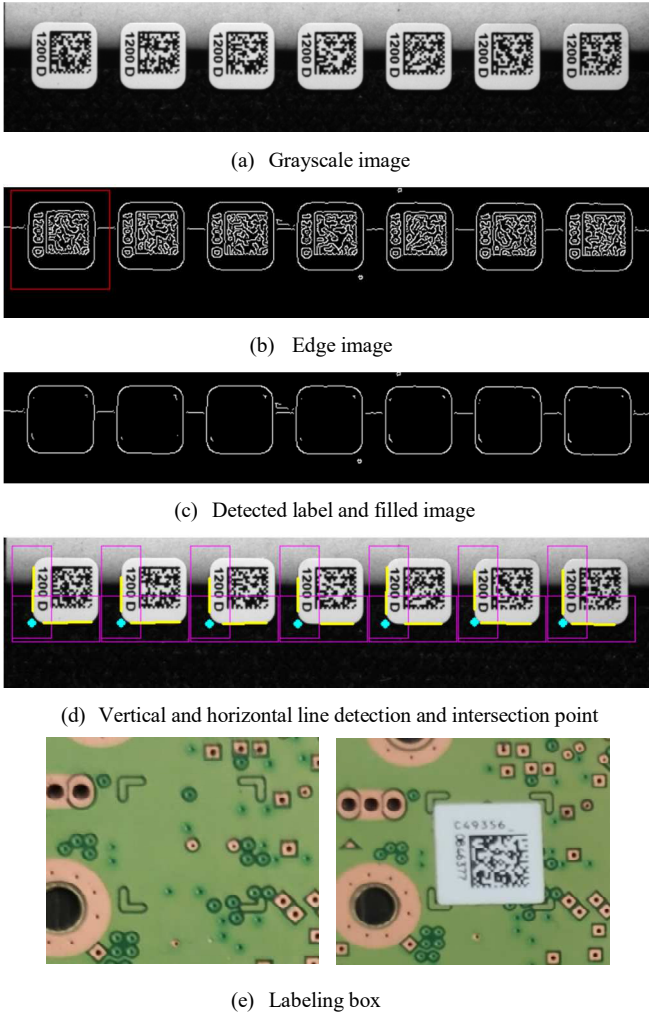


Fig. 3. a) Grayscale image, b) Edge image, c) Detected label & filled image, d) Vertical and horizontal line detection and intersection points e) Labeling box, All images shown are only a part of the full image.

$$x = \frac{\begin{vmatrix} x_1 & y_1 & x_1 - x_2 \\ x_2 & y_2 & x_1 - x_2 \\ x_3 & y_3 & x_3 - x_4 \\ x_4 & y_4 & x_3 - x_4 \end{vmatrix}}{\begin{vmatrix} x_1 - x_2 & y_1 - y_2 \\ x_3 - x_4 & y_3 - y_4 \end{vmatrix}}$$

$$y = \frac{\begin{vmatrix} x_1 & y_1 & y_1 - y_2 \\ x_2 & y_2 & y_1 - y_2 \\ x_3 & y_3 & y_3 - y_4 \\ x_4 & y_4 & y_3 - y_4 \end{vmatrix}}{\begin{vmatrix} x_1 - x_2 & y_1 - y_2 \\ x_3 - x_4 & y_3 - y_4 \end{vmatrix}} \quad (1)$$

$$Angle = \arctan\left(\frac{y_3 - y_4}{x_3 - x_4}\right) \quad (2)$$

C. Image and Robot Coordinate transformation

A perspective Transformation [17] is to reduce the image tilting and distortion caused by the 3D world position obtained by the camera to 2D plane coordinates [18]. The transformation matrix is equation (3).

$$\begin{bmatrix} x_p \\ y_p \\ 1 \end{bmatrix} = M \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \quad (3)$$

where the transform matrix, M see equation (4), was derived from the four center points of circles in the image calibration.

$$M = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} \quad (4)$$

To calculate the conversion matrix, first a Canny edge detector was used to find pattern edges and Hough transform, found the circle centers – see the four points, P1, P2, P3, and P4, in the calibration image – seen figure (5). Using the OpenCv getPerspectiveTransform function to compute the M matrix using the steps in figure (4).

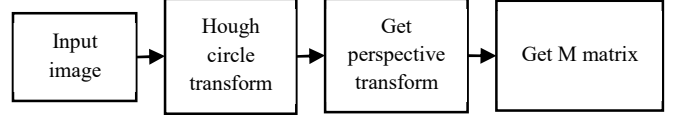


Fig. 4. Calibration and coordinate conversion

After obtaining the M matrix, the target intersection (x_p, y_p) for the robot was calculated by computing the determinant the equation (3), thus the result is an equation (5):

$$x_p = \frac{m_{11}x + m_{12}y + m_{13}}{m_{31}x + m_{32}y + m_{33}}$$

$$y_p = \frac{m_{21}x + m_{22}y + m_{23}}{m_{31}x + m_{32}y + m_{33}} \quad (5)$$

Table I show the result of a calibration and conversion matrix for converting pixel position to the world coordinates (mm) using equation (5).

TABLE I. FOUR-POINTS CORRESPONDING AND RESULTANT M

Coordinate Location	Image (pixel)		World (mm)	
	x	y	x'	y'
P1	1087	1152	132.86	-283.56
P2	1422	1150	132.67	-255.98
P3	1085	981	118.72	-283.42
P4	1421	981	118.82	-255.98
M	$\begin{bmatrix} 0.00403 & 0.09423 & 30.63460 \\ 0.08097 & -0.01092 & -381.8564 \\ 0.00003 & 0.00004 & 1 \end{bmatrix}$			

D. Experiments results

In the experiment, we captured 12 rows of different label images with various characters as shown in figure (6) to test performance. Each label image was tested 30 times. We evaluated the results by measuring the actual positioning and rotation angle of robot coordinates with the computed positions of image coordinates in each row of labels using equation (6).

$$Errorr = |P_{robot} - P_{computed}| \quad (6)$$

where P_{robot} is an actual robot point and $P_{computed}$ is a computation was measured from image coordinate. The tests were run with and without the additional light, using the ambient laboratory fluorescents alone.

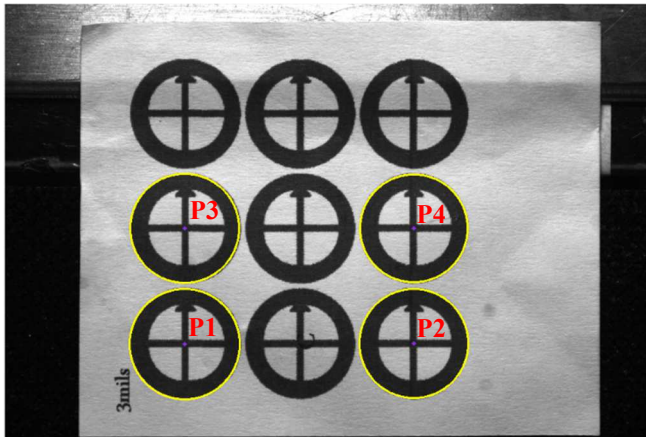


Fig. 5. Calibration and coordinate conversion Image

TABLE II show the average, minimum and maximum positioning error for label position 1 to 7. The overall of coordinate error sorted in order (x, y, angle) where the average error was (0.11 mm, 0.07 mm, 0.90 °), the minimum error was (0.01 mm, 0.01 mm, 0.05 °) and the maximum error was (0.25 mm, 0.32 mm, 3.94 °).

TABLE III shows the result of each position of labels. With the light source, all label position was detected with a 100%. The detected rate without the light source, label positions 1 to 4 was 83% and the following label positions 5, 6 and 7 were 75%, 41% and 25%. The computed time of seven position of labels including the edge detection, finding straight line and coordinate transforming was ~180 ms.

For this robot, the vacuum head was 5.5×5.5 mm in size, whereas the targets where 6×6 mm, so an acceptable tolerance was 0.5 mm. Hence, our results show that the robot should be able to grip the label every time.



Fig. 6. Partial experiment results with the rotation angle of label

IV. CONCLUSION AND FUTURE WORK

The system provided sufficiently accurate picking coordinates for the SCARA robot to grip the label effectively. Here, we developed the label attachment machine which focused on the picking point. Our method relied on straight line detection to locate a picking point and to position the 6 mm diameter labels in a machine system integrated into a SCARA robot. Moreover, a perspective transform allowed

labels to be rotated relative to the camera axis. In addition, the defined four points of the robot are important for transforming coordinates in order to achieve accuracy. Through the factory environment, the proper light source is also important for the proposed system.

TABLE II. THE EXPERIMENT RESULT OF POSITIONING ERROR

Position	Error	x-error (mm)	y-error (mm)	Angle-error (degree)
1	AVERAGE	0.11	0.04	0.92
	MIN	0.03	0.01	0.09
	MAX	0.18	0.08	3.88
2	AVERAGE	0.07	0.07	1.41
	MIN	0.01	0.02	0.09
	MAX	0.19	0.17	3.94
3	AVERAGE	0.09	0.10	0.83
	MIN	0.01	0.03	0.06
	MAX	0.22	0.27	1.43
4	AVERAGE	0.09	0.04	0.88
	MIN	0.01	0.02	0.06
	MAX	0.16	0.12	1.22
5	AVERAGE	0.09	0.07	0.76
	MIN	0.01	0.01	0.06
	MAX	0.25	0.29	2.60
6	AVERAGE	0.13	0.12	0.75
	MIN	0.02	0.02	0.05
	MAX	0.23	0.29	2.15
7	AVERAGE	0.16	0.09	0.78
	MIN	0.09	0.02	0.06
	MAX	0.23	0.92	2.39
AVERAGE		0.11	0.07	0.90
MIN		0.01	0.01	0.05
MAX		0.25	0.32	3.94

TABLE III. DETECTED LABEL WITH AND WIHTOUT THE LIGHT SOURCE

Label Position	Detected rate		Computed time (ms)
	with light (%)	without light (%)	
1	100	83.33	~180
2	100	83.33	
3	100	83.33	
4	100	83.33	
5	100	75.00	
6	100	41.67	
7	100	25.00	

For future study, a conveyor tracking will be studied to upgrade the system at the robot placing side, after gripping the label, the robot could be attached the label to the PCB while it is moving on the conveyor with the various model changing automatically.

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